

# A Reproducible GeoAI Framework for Explainable Spatiotemporal Analysis of National Missing Persons Records: Multi-Algorithm Clustering and Multi-Level Validation (Peru, 2019–2025)

Moisés Evangelista Gamarra<sup>1</sup>, Edwar Abraham Alama Carreño<sup>1,\*</sup>

<sup>1</sup> Servicio Nacional de Adiestramiento en Trabajo Industrial (SENATI), Lima, Peru

\* Corresponding author: Edwar Abraham Alama Carreño. E-mail addresses: 1634030@senati.pe (E.A. Alama Carreño, corresponding author), mevangelistag@senati.pe (M. Evangelista Gamarra).

## Highlights

- A national missing-persons registry yields a reproducible two-cluster partition.
- One feature, departmental report share, drives 82% of the separating variance.
- Removing this feature yields a markedly different partition (ARI = -0.05 vs. original).
- The apparent socioeconomic association is an artifact of the wrong unit of analysis.
- The partition remains highly stable when applied to the later 2025 evaluation year.

## Abstract

This study asks whether an unsupervised clustering pipeline applied to Peru's national missing-persons registry (RENIPED, N = 77,331 records, 2019-2025) recovers a genuine multidimensional territorial-temporal structure, or mainly rediscovers which department reports the most cases. Six clustering algorithms are compared using 2019-2024 for model development and 2025 for evaluation, and the choice between two and three clusters is resolved through eight explicit, tabulated criteria rather than a single validity score. The two-cluster solution, which separates Lima Metropolitana from the rest of the country, is adopted as the primary model. A dedicated ablation shows why: one engineered feature, the department's share of all reports (DPTO\_FREQ), accounts for 82% of the variance that separates the two clusters, and removing it yields a markedly different partition (adjusted Rand index, ARI, near zero against the original). A four-covariate socioeconomic comparison initially appeared to validate the clusters ( $p < 0.001$ ) but is shown to be a pseudoreplication artifact: at the correct department-level unit of analysis (25 departments, not 65,873 repeated rows) none of the four associations remain significant. Applied without refitting to 2025, the model remains highly stable (ARI = 1.0000); because one territorial feature (DPTO\_FREQ) was built from the full 2019-2025 span, this is evidence of partition stability rather than a fully independent temporal validation. Taken together, the results support a narrow claim: RENIPED contains a territorial partition recovered consistently across six algorithms, but that partition reflects reporting concentration, not a socioeconomically validated or individually predictive risk structure.

**Keywords:** *GeoAI; missing persons; multi-algorithm clustering; feature dominance; pseudoreplication; reproducibility; Peru*

## 1. Introduction

Missing-persons reporting is a recurring humanitarian and public-safety problem worldwide. The International Committee of the Red Cross lists tracing the missing among its core strategic priorities for 2024-2027 (ICRC, 2024), and the International Commission on Missing Persons documents tens of thousands of open cases each year across conflict, migration and criminal contexts (ICMP, 2024). Peru is a case where this problem is both large and visibly worsening: the National Registry of Information on Missing Persons (RENIPED), maintained by the National Police, is the country's primary administrative source for these events (Peruvian National Police, n.d.), and reporting through the most recently published period indicates the upward trend has continued (Quinteros, 2026). Civil-society organizations, the national Ombudsman's Office and investigative press have independently converged on the same observation: disappearances associated with gender-based violence have risen sharply since 2023 (NACLA Report on the Americas, 2025; Infobae, 2025a; Infobae, 2025b; Amnesty International Peru, 2025; Quinteros, 2025; Peruvian Ombudsman's Office, 2025) — consistent with a broader regional pattern of high and rising gender-based violence across Latin America and the Caribbean (World Bank, 2022).

A registry of this kind is normally read descriptively: how many cases, in which places, involving whom. That reading cannot show whether cases cluster into recurring situational profiles, whether those profiles are stable across time and space, or whether an apparent pattern is a real multidimensional structure rather than a restatement of one easy-to-compute quantity, such as which department reports the most cases. Answering those questions requires treating the registry as a spatiotemporal dataset and applying methods designed to recover structure without a predefined label — which is precisely the role that geospatial artificial intelligence, or GeoAI (the application of machine-learning methods to explicitly spatial problems), has been proposed to play in administrative and public-safety data (Li, 2020; Janowicz et al., 2020; Li et al., 2024).

Unsupervised clustering is the most direct GeoAI tool for this purpose, but it carries a specific and under-discussed risk when applied to administrative counts: because there is no ground-truth label to check a clustering result against, a partition can look like a scientific discovery when it is, in fact, a restatement of whichever single variable is easiest to compute, most repeated across records, or most strongly correlated with population size. This risk is not hypothetical for missing-persons and crime data, where case counts are mechanically concentrated in the largest cities. A clustering pipeline that does not explicitly test for this possibility cannot distinguish a genuine multidimensional territorial-temporal structure from a single-variable artifact dressed up as one — and, if the distinction is not made, risks being read as evidence of something the data cannot support, such as an individual-level risk profile.

A second, methodological gap follows. Standard practice offers no agreed procedure for choosing between two close candidate solutions (two clusters versus three, say), for telling a real external validation apart from one that pseudoreplicates a handful of departmental values across thousands of rows, or for confirming that a post-hoc explainability method describes the clustering itself rather than a separate model built to imitate it. Each of these is a known failure mode in the broader unsupervised-learning literature, but they are rarely tested together, explicitly, inside a single reproducible pipeline applied to real administrative data.

This concern is condensed here into three explicit research questions. **RQ1.** Does the clustering framework recover a reproducible multidimensional structure, or does it mainly reproduce territorial reporting concentration? **RQ2.** How sensitive is the inferred partition to the territorial-frequency information built into the feature set? **RQ3.** Does the observed structure remain recoverable across spatial aggregation and an unseen year?

This paper re-executes, from a freshly downloaded copy of RENIPED, a national-scale clustering pipeline, testing rather than assuming its own prior conclusions. It treats RQ1–RQ3 above, together with the further methodological concerns raised by that review, as testable questions and reports what a full re-analysis answers: (i) whether the clustering result is a genuine multidimensional structure or the signature of a single territorial-frequency feature; (ii) whether six algorithms converging on the same partition constitutes independent evidence for that partition's cluster count, given that the count itself was fixed before the algorithms were compared; (iii) whether a socioeconomic "validation" of the clusters survives correction for pseudoreplication; and (iv) whether an explainability step built on a substitute model can be presented as explaining the clustering directly. The contribution is accordingly methodological and territorial rather than algorithmic: a worked, disclosed template for closing exactly this class of question with reproducible evidence — quantifying feature dominance directly rather than asserting its absence, treating a close model-selection decision as a multi-criteria trade-off rather than a single number, and distinguishing a genuine unit-of-analysis error from a superficially similar but valid aggregate finding — applied here to Peru's national missing-persons registry, and potentially transferable to other administrative registries with comparable territorial concentration.

## 2. Related Work

Quantitative research on missing-persons data falls into two groups that rarely overlap. The first, larger group treats the problem descriptively or predictively at the level of the individual case: national profiling studies (James et al., 2008), trajectory analyses of runaway youth (Jeanis et al., 2020; Pediatrics, 2020), and ecological studies of the demographic correlates of violence against adolescents in Peru specifically (Enríquez et al., 2024). These studies establish which individual and family-level factors co-occur with disappearance risk, but by design they say nothing about whether cases cluster into recurring territorial-temporal profiles once aggregated into a registry, because that question is outside their unit of analysis. A recent systematic review of the field confirms this gap directly, finding that spatial clustering approaches to missing-persons data remain comparatively underexplored relative to descriptive and inferential statistics (Ruiz Reyes et al., 2025).

The second, smaller group applies clustering or machine learning to missing-persons registries directly, but for a narrower operational purpose than characterizing registry-wide structure. Johnson et al. (2025) is the closest prior application of spatial clustering to missing-persons data, but it clusters cases to recommend community-outreach event locations, not to characterize national-level territorial-temporal structure or to test whether that structure survives removal of the variable that produces it. Pierzchała and Gutowski (2021) apply machine learning to a related but distinct predictive task, recommending a case's "search level," which presupposes rather than tests cluster validity. Neither line of work asks the question this paper treats as central: whether an unsupervised partition of an administrative registry is a genuine multidimensional pattern or the recoverable signature of one dominant, easily computed feature — and, if the latter, what that implies for how the partition should and should not be interpreted.

This gap has a direct methodological consequence. Because no prior missing-persons clustering study reports a feature-ablation test, a null-adjusted consensus metric, or a pseudoreplication-corrected external validation side by side with the uncorrected version, there is no existing template a practitioner can point to for defending — or, as this paper does, for correcting — exactly these choices in a registry-clustering pipeline. The tools required to close that gap are not novel in

isolation: internal cluster-validity indices (Silhouette: Rousseeuw, 1987; Davies-Bouldin: Davies and Bouldin, 1979; Calinski-Harabasz: Caliński and Harabasz, 1974), partition-agreement statistics (Hubert and Arabie, 1985), resampling-based stability estimates (Efron, 1979), and spatial-autocorrelation diagnostics (Moran, 1950; Anselin, 1995; Getis and Ord, 1992; Ord and Getis, 1995) all have a long track record in spatial epidemiology and other GeoAI applications (Auchincloss et al., 2012; VoPham et al., 2018). What is missing from the missing-persons literature specifically is their joint, disclosed application to a single registry-scale clustering decision, together with an explicit test of whether a post-hoc explainability method (Lundberg and Lee, 2017, applied here to a Random Forest surrogate: Breiman, 2001) is describing the clustering itself or a separate model trained to imitate it. Administrative missing-persons records also carry well-documented data-quality and case-classification concerns in their own right (Ferguson and Picknell, 2021), which is a second, independent reason for the transparency this paper applies to preprocessing (Section 3.2) rather than treating the registry export as ready-made analytical input.

This paper closes that gap for one national registry by pairing that same standard toolkit with two checks that are specific to registry data: an ablation test of the territorial-frequency feature that a raw case count mechanically produces, and a re-derivation of any covariate-level "validation" at the unit of analysis the data actually supports, rather than at the unit of analysis that happens to have the most rows.

### 3. Materials and Methods

#### 3.1 Data source and study population

The study population is every record in RENIPED, the national missing-persons registry maintained by the Peruvian National Police (Peruvian National Police, n.d.), reported between January 2019 and December 2025 ( $N = 77,331$ ). Each row is one report and carries department, province and district of occurrence, sex, a six-level age bracket, nationality, year, month and a record-count field; no name or other direct identifier is present in the export used. The export was re-downloaded specifically for this re-analysis and its content hash was verified, byte-for-byte, against every prior version of this pipeline, so the numbers below are not carried over from an earlier run. Peru's 26 departments are all represented, but unevenly: Lima Metropolitana alone accounts for 23.0% of training-period records, roughly three times the next largest department (Figure 2) — a concentration this paper treats throughout as a variable to be tested, not assumed away.

#### 3.2 Data quality assessment and preprocessing

Administrative missing-persons records carry well-documented data-quality and case-classification concerns in general (Ferguson and Picknell, 2021), so every correction applied here is disclosed rather than absorbed silently into later steps; Table 1 documents each one, the records it affected, and how it was verified. Two corrections are worth highlighting in the text. First, sex and age bracket had missing entries in the raw export (0.1% and 1.5% of records respectively), resolved through department-conditional modal imputation for age bracket and dataset-wide modal imputation for sex. Second, the text-to-ordinal encoding of age bracket inherited a bug from an earlier pipeline version that matched age ranges via text substring and miscoded 18.4% of records into the wrong bracket; the corrected and uncorrected encodings happen to produce an identical final  $K = 2$  partition ( $ARI = 1.0000$  between them), so the bug does not change the final  $K = 2$  partition, but it is corrected here because other age-based analyses (Section 4) use the ordinal values directly.

**Table 1. Data quality assessment and preprocessing corrections applied to the raw RENIPED export (N = 77,331), each verified against the executed pipeline rather than assumed (Section 3.2).**

| Issue identified                                                                       | Affected records                                | Correction applied                                                                                                     | Verification                                                                                |
|----------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Duplicate rows                                                                         | 0                                               | None required                                                                                                          | Confirmed via exact row-duplicate check                                                     |
| Missing sex (SEXO)                                                                     | 83 (0.1%)                                       | Filled with the dataset-wide modal category                                                                            | 0 missing after correction                                                                  |
| Missing age bracket (RANGO_EDAD)                                                       | 1,197 (1.5%)                                    | Filled with the department-level modal category; dataset-wide mode used only where no departmental value was available | 0 missing after correction                                                                  |
| Inconsistent category spelling/case                                                    | All categorical fields                          | Whitespace-stripped and upper-cased before grouping                                                                    | Verified against the six known age-bracket labels                                           |
| Age-bracket text-to-ordinal miscoding (bug inherited from an earlier pipeline version) | 18.4% of records                                | Replaced substring matching with exact-label matching plus a numeric-extraction fallback                               | Corrected and uncorrected encodings yield an identical final K = 2 partition (ARI = 1.0000) |
| Extreme/invalid record counts (CANTIDAD)                                               | Below 1st or above 99th percentile, or negative | Clipped to the 1st-99th percentile range; negatives replaced with the median                                           | Prevents a small number of extreme values from dominating volume-derived features           |

### 3.3 Variable construction and feature engineering

Eleven features were constructed per record for clustering, grouped here into five conceptual families rather than an undifferentiated list (Table 2 gives the definition, scale, and methodological risk of each). A demographic family contributes an ordinal age bracket (EDAD\_ORD). A territorial-concentration family contributes the single most consequential variable in this study, DPTO\_FREQ — the share of all 77,331 records, across the full 2019-2025 span, occurring in the same department as the current record: a direct, mechanical measure of how much of the national count one department contributes, which is exactly why Section 4.4 tests, rather than assumes, how much of the clustering result it is responsible for. Because it is computed from the full study period rather than the training window alone, DPTO\_FREQ does not achieve complete temporal isolation between training and the 2025 test period; Section 5.6 treats this as an open limitation and moderates the Section 4.9 generalization claim accordingly. A volume family contributes a log record count (LOG\_CANTIDAD) and a per-capita log rate (LOG\_RATE\_10K), separating how many cases a department reports from how many it reports relative to its size — a distinction DPTO\_FREQ alone cannot make. A temporal family contributes a normalized year, a four-level COVID-19 phase indicator motivated by regional evidence that the pandemic disrupted gender-based-violence incidence and reporting alike (Carvajal, 2020; Perez-Vincent and Carreras, 2022), and a linear time index. A seasonality-and-anomaly family contributes a cyclical month encoding, a 12-month rolling mean, and a department-month anomaly z-score. All eleven features were scaled with a robust (median/interquartile-range) scaler fit on the training partition only.

**Table 2. Dictionary of the eleven engineered features used for clustering, grouped by conceptual family (Section 3.3).**

| Family                    | Variable         | Definition                                                                                                          | Scale               | Analytical role                              | Main methodological risk                                                                                      |
|---------------------------|------------------|---------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Demographic               | EDAD_ORD         | Age bracket recoded to an ordinal scale (0 = 0-5 years ... 5 = 60+ years)                                           | Ordinal, 0-5        | Age-context feature                          | Coarse 6-level bracket may mask within-bracket heterogeneity                                                  |
| Territorial concentration | DPTO_FREQ        | Share of all national records (2019-2025) occurring in the same department as the record                            | Continuous, 0-1     | Central variable of this study (Section 4.4) | Dominates the K=2 solution by construction; built from the full study period, not training-only (Section 5.6) |
| Volume                    | LOG_CANTIDAD     | Natural log of the record-count field (log1p)                                                                       | Continuous          | Raw report volume, right-skew reduced        | Correlated with DPTO_FREQ and with population size                                                            |
| Volume, per-capita        | LOG_RATE_10K     | Log of reports per 10,000 inhabitants, using a fixed department population figure                                   | Continuous          | Separates volume from population size        | Population figure is a single fixed cross-section, not year-specific                                          |
| Temporal                  | ANNO_NORM        | Year linearly rescaled to 0-1 over 2019-2025                                                                        | Continuous, 0-1     | Coarse time trend                            | Linear only; cannot capture non-monotonic year effects                                                        |
| Temporal                  | COVID_PHASE      | Four-level indicator: pre-pandemic (2019) / acute (2020-2021) / recovery (2022-2024) / post (2025)                  | Ordinal, 0-3        | Tests pandemic phase vs. calendar time       | Phase boundaries are calendar-based, not epidemiologically validated                                          |
| Temporal                  | TREND_INDEX      | Linear month index from the first to the last month in the data, rescaled to 0-1                                    | Continuous, 0-1     | Fine-grained time trend                      | Collinear with ANNO_NORM at the annual scale                                                                  |
| Seasonality               | MES_SEN, MES_COS | Sine and cosine of calendar month                                                                                   | Continuous, -1 to 1 | Cyclical month encoding                      | Assumes one symmetric annual cycle                                                                            |
| Anomaly                   | MONTHLY_ANOMALY  | Department-month z-score of LOG_CANTIDAD vs. that department's typical value for the same month, clipped to $\pm 3$ | Continuous          | Flags department- and season-specific spikes | Denominator uses a small-sample standard deviation; clipping bounds but does not remove this                  |
| Trend/ momentum           | ROLLING_MEAN_12  | 12-month rolling mean of LOG_CANTIDAD within department                                                             | Continuous          | Local temporal momentum                      | Introduces limited look-back dependency on a department's own recent history                                  |

### 3.4 Temporal experimental design

All model selection, sensitivity analysis and explainability work in this paper was performed on a training partition restricted to 2019-2024 ( $N = 65,873$ ); the full 2025 year ( $N = 11,458$ ) was set aside and used exactly once, for the final generalization check reported in Section 4.8. This partitioning avoids a leakage pathway in which a department-year aggregation computed over the full 2019-2025 span would be partly reused as a temporal validation set — meaning the "test" would have already seen part of what it was testing.

### 3.5 Clustering algorithms

Six clustering algorithms, spanning centroid-based (K-Means, Mini-Batch K-Means), hierarchical (Ward-linkage Agglomerative Clustering), probabilistic (Gaussian Mixture Models), tree/incremental (BIRCH, which builds a bounded

clustering-feature tree rather than using a density-reachability criterion) and graph-based (Spectral Clustering, nearest-neighbor affinity) families, were fit to the same scaled training features using their scikit-learn implementations (Pedregosa et al., 2011). No single algorithm's assumptions (spherical clusters, Gaussian components, connectivity) match this problem a priori — which is the point of comparing six: a partition that appears only under one algorithm's specific assumptions is evidence of an assumption artifact, not a data pattern. Agreement across the six families is therefore treated as one input to the model-selection decision (Section 3.6, Table 4) — a source of increased confidence in a partition's stability, not, by itself, an independent validation of how many clusters that partition should have, since the candidate cluster counts compared were fixed before the algorithms were run (Section 4.2 returns to this distinction explicitly).

**Table 3. The six clustering algorithms compared in this study, their structural family, and the assumption each introduces (Section 3.5).**

| Algorithm                     | Family           | Structural assumption                                                                 | Configuration in this pipeline   |
|-------------------------------|------------------|---------------------------------------------------------------------------------------|----------------------------------|
| K-Means                       | Centroid-based   | Spherical, similarly sized clusters around K centroids                                | n_init=10, fixed seed (42)       |
| Mini-Batch K-Means            | Centroid-based   | Same as K-Means, fit via random mini-batches                                          | n_init=10, fixed seed            |
| Ward Agglomerative Clustering | Hierarchical     | Bottom-up merging that minimizes the increase in within-cluster variance              | Ward linkage, Euclidean distance |
| Gaussian Mixture Model        | Probabilistic    | Data generated by a mixture of K Gaussian components (allows overlap)                 | n_init=5, fixed seed             |
| BIRCH                         | Tree/incremental | Bounded clustering-feature (CF) tree built incrementally, then merged into K clusters | scikit-learn default threshold   |
| Spectral Clustering           | Graph-based      | Clusters via eigenvectors of a nearest-neighbor similarity graph                      | 15-neighbor affinity, fixed seed |

### 3.6 Model selection

Choosing between a small number of plausible cluster counts is a known open problem in unsupervised learning: no single criterion is authoritative, and criteria frequently disagree with one another. This paper does not resolve that disagreement by picking the criterion that happens to favor one answer; instead, eight criteria were computed for  $K = 2$  through  $K = 10$  (or, where a criterion is only meaningful as a pairwise comparison, for the two leading candidates) and compared side by side in a single decision table (Table 4) that states in plain language what each one measures, so the trade-off — not just the final choice — is auditable. The internal-validity, agreement, stability, spatial-coherence, feature-dominance and generalization criteria follow standard definitions (Rousseeuw, 1987; Davies and Bouldin, 1979; Caliński and Harabasz, 1974; Hubert and Arabie, 1985; Tibshirani et al., 2001); the Consensus\_Score bipolarity index is defined for this study specifically, not an externally established metric. It measures, for every pair of records, whether the six algorithms agree they belong to the same cluster or agree they belong to different clusters, and averages that agreement across all pairs, so that either kind of strong agreement counts toward the score; because such a score rises somewhat mechanically with  $K$  even under random labels, it is reported net of its own null-simulation baseline and weighted no more heavily than the other seven criteria.

### 3.7 Spatial validation

Department-level global spatial autocorrelation (Moran's I, Geary's C, Getis-Ord G) was computed on a  $K = 4$  nearest-neighbor weight matrix over department-capital coordinates, for three variables: the minority-cluster (Lima-associated) share, DPTO\_FREQ, and raw case volume. Spatial block cross-validation additionally partitioned the 26 departments into six geographic blocks (via K-Means on capital coordinates) and recorded, for each block, whether it contained records from both clusters — a block containing only one cluster cannot support a within-block agreement estimate.

### 3.8 Robustness and sensitivity analysis

Three sensitivity families were tested. Feature-dominance sensitivity re-clustered the training data at  $K = 2$  after progressively removing candidate territorial and volume features (DPTO\_FREQ alone; DPTO\_FREQ and

LOG\_RATE\_10K together; a four-feature minimal set with no territorial variable at all) and compared each reduced-feature partition against the full-feature final partition by adjusted Rand index. Lima-Metropolitana sensitivity re-clustered the data after excluding Lima, and again after additionally excluding the contiguous, functionally integrated department of Callao, each time compared by adjusted Rand index against a  $K = 3$  reference partition ( $K = 3$  rather than  $K = 2$  is used only for this specific comparison, because restricting the  $K = 2$  partition's own labels to non-Lima rows leaves a single constant value, against which agreement is undefined). Multiscale aggregation sensitivity independently re-clustered the data collapsed to department-year ( $N = 156$ ) and to department-month ( $N = 1,864$  of 1,872 theoretically possible) cells, to test whether the individual-record partition is recoverable from purely territorial-temporal aggregates that no longer contain any single record's information.

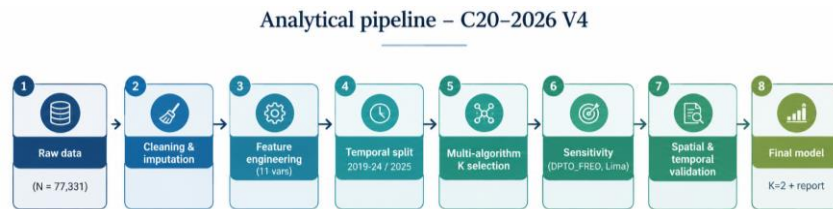
### 3.9 Explainability using a surrogate model

K-Means and the other algorithms in Section 3.5 have no native, model-intrinsic mechanism for attributing a clustering decision to individual features, unlike a supervised classifier. To obtain feature attributions at all, this paper follows a standard two-step substitute-model procedure: a Random Forest classifier (Breiman, 2001) is trained, on a 75/25 stratified split, to predict each record's already-assigned cluster label from the same eleven features used to build that partition; SHAP values (Lundberg and Lee, 2017) are then computed for that Random Forest, on the held-out 25%, via TreeExplainer. This chain matters for interpretation: SHAP explains why the Random Forest reproduces a given label, not why K-Means originally assigned it, and the two are the same only to the extent the surrogate's near-perfect fidelity (it reproduces the cluster label with 100% accuracy for  $K = 2$ , and 99.95% for the  $K = 3$  surrogate used in Section 4.2) implies. Near-perfect fidelity in this specific setting is read here as a symptom of a low-dimensional, feature-dominated partition that is easy for any classifier to reproduce, not as evidence that the explanation is unusually trustworthy. Permutation importance on the same surrogate is reported alongside SHAP as a second, methodologically independent attribution check, precisely because the two methods can and do disagree (Section 4.2).

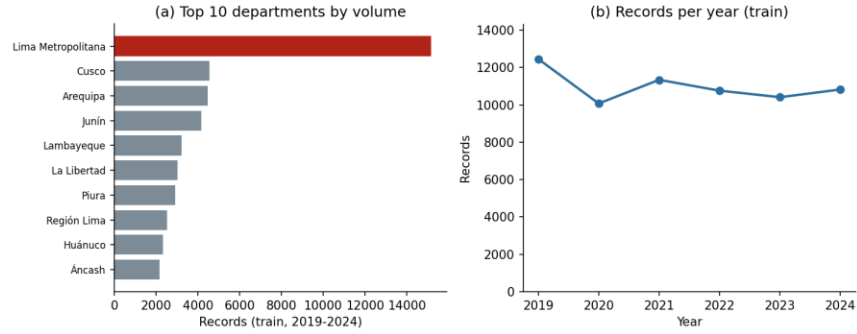
## 4. Results

### 4.1 What does the dataset look like, and where does the concentration this study investigates come from?

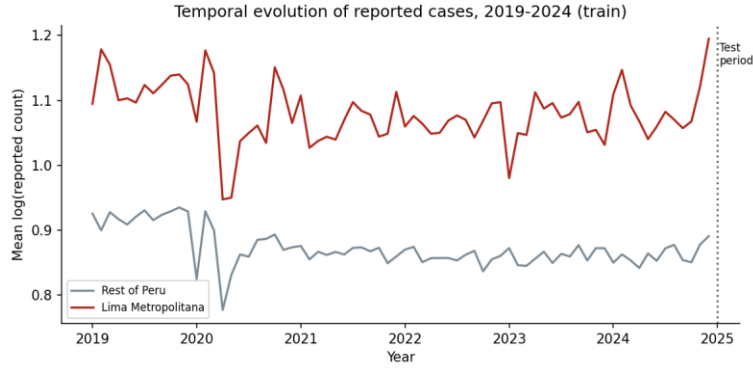
After cleaning, the training partition (2019-2024) contains 65,873 records across 26 departments (54.85% recorded as female); the 2025 test partition contains 11,458 records. Lima Metropolitana accounts for 23.0% of training records (15,143 of 65,873) and 23.1% of the full 2019-2025 dataset (17,829 of 77,331) — the two figures differ only because they are computed over different denominators, not because of any inconsistency in the underlying counts. Figure 1 summarizes the full analytical pipeline; Figure 2 shows that this concentration is not a one-year anomaly (10,075-12,459 records per year, no single year responsible) and that Lima's share is roughly three times that of the next-largest department, Cusco. Figure 8 adds the temporal dimension: Lima Metropolitana and the rest of the country follow visibly different but individually stable monthly trajectories throughout 2019-2024, with no structural break at the COVID-19 phase boundaries used to build the COVID\_PHASE feature (Section 3.3) — which is why phase is treated as a covariate rather than as a segmentation point for the clustering itself. This is the concentration that Sections 4.4 and 4.5 test directly, rather than treat as an assumed background fact.



**Fig. 1.** The six-stage analytical pipeline followed in this study, from the raw registry export through to the final validated model; each stage is described in the correspondingly numbered subsection of Section 3.



**Fig. 2. (a) The ten departments with the most training-period reports; Lima Metropolitana (highlighted) reports roughly three times as many as the next-largest department, Cusco. (b) Total reports per year, 2019-2024: year-to-year variation is modest (10,075-12,459 records) and no single year drives the department-level concentration shown in (a).**

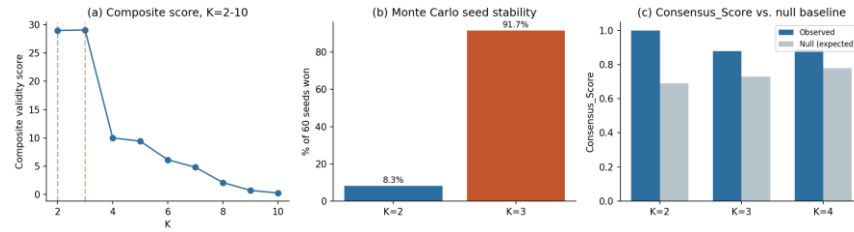


**Fig. 8. Mean log(reported count) per month, Lima Metropolitana versus the rest of Peru, training period 2019-2024 (dashed line marks the start of the held-out 2025 test period). The two series follow visibly different but individually stable trajectories, with no structural break at the COVID-19 phase boundaries used to construct the COVID\_PHASE feature (Section 3.3).**

#### 4.2 Which clustering structure is supported, and how is the choice between two and three clusters actually made?

The eight criteria from Section 3.6 do not unanimously favor one candidate (Table 4): three clearly favor  $K = 2$  (algorithm-family stability, Consensus\_Score net of null, and spatial coherence of the extra cluster), one clearly favors  $K = 3$  (Monte Carlo seed stability, at 91.7% of 60 resampled seeds under the design detailed in Section 3.6, sharpening to 100% under a more thorough resampling variant), two are effectively tied (composite validity, 28.96 vs. 29.06; and temporal generalization,  $ARI = 1.0000$  vs. 0.9851), the Gap Statistic is uninformative either way, and feature-dominance resistance favors  $K = 3$  numerically (its top feature, again DPTO\_FREQ, accounts for 40.2% of eta-squared mass, with three further features — LOG\_RATE\_10K, MONTHLY\_ANOMALY and LOG\_CANTIDAD — each adding a further 15-28%, versus 82.0% concentrated in DPTO\_FREQ alone at  $K = 2$ ) but is qualified by spatial coherence:  $K = 3$ 's additional cluster is the majority label in only 2 of 26 departments (Madre de Dios and Moquegua, both low-population) and is otherwise scattered by month, defined chiefly by an unusually high MONTHLY\_ANOMALY (mean 1.73, versus -0.27 for the other non-Lima group and -0.01 for Lima itself), the profile of a temporal-volatility episode rather than a persistent geography, so a feature-diverse but spatially scattered cluster is not obviously preferable to a feature-dominated, spatially coherent one.

$K = 2$  is therefore selected as the primary model;  $K = 3$  is retained as a secondary finding, interpreted as a temporal-volatility regime rather than an unsupported third territorial category. A SHAP surrogate computed for  $K = 3$  specifically to test this interpretation identifies the same two features already implicated by the variance analysis, DPTO\_FREQ and LOG\_RATE\_10K (their order reversed between SHAP and permutation importance) — a confirmatory check that does not change the  $K = 2$  selection.



**Fig. 3.** Three of the eight model-selection criteria from Table 4, computed on the training partition. (a) The composite internal-validity score (Section 3.6) across  $K = 2$ -10:  $K = 2$  and  $K = 3$  are effectively tied, and both outperform  $K \geq 4$ . (b) Percentage of 60 resampled seeds in which each  $K$  maximizes Silhouette:  $K = 3$  wins more often under this specific criterion. (c) The Consensus\_Score bipolarity index (solid bars) against its own null-simulation baseline (light bars) at  $K = 2, 3$  and 4:  $K = 2$ 's advantage over its null is the largest of the three.

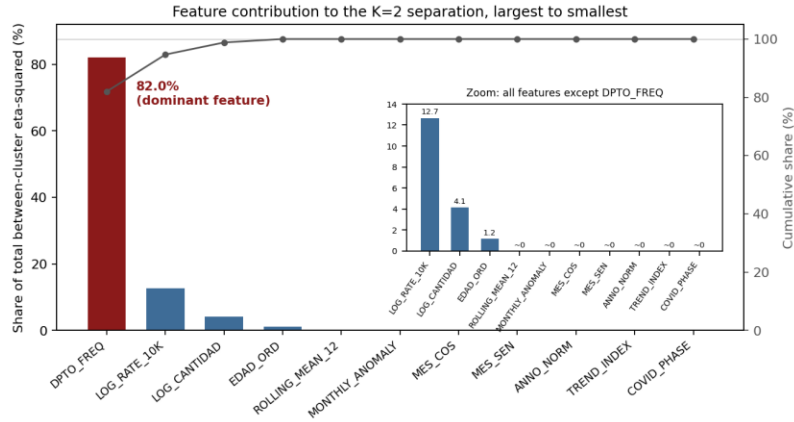
**Table 4.** Final model-selection decision: eight criteria compared for  $K = 2$  vs.  $K = 3$ , each with what it measures and how it was interpreted.  $K = 2$  is selected as the primary model;  $K = 3$  is retained as a secondary finding.

| Criterion                                                            | What it measures                                                                               | $K = 2$                                      | $K = 3$                                                                  | Preferred                | Interpretation                                                                                           |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------|
| Composite validity (Silhouette + Davies-Bouldin + Calinski-Harabasz) | Combines three standard internal-validity indices into one 0-30 scale (Section 3.6)            | 28.96<br>(Sil=0.4496, DBI=0.9235, CH=41,817) | 29.06<br>(Sil=0.4546, DBI=0.9622, CH=43,298)                             | Tie<br>( $\Delta=0.10$ ) | Neither favored; both clearly outperform $K \geq 4$                                                      |
| Algorithm-family stability                                           | Pairwise ARI among the six clustering algorithms (Section 3.5)                                 | 1.000<br>(near-unanimous)                    | 0.664<br>(moderate)                                                      | $K = 2$                  | Six algorithmic assumptions converge far more consistently at $K = 2$                                    |
| Monte Carlo seed stability                                           | % of 60 resampled runs in which each $K$ maximizes Silhouette                                  | 8.3% of seeds                                | 91.7% of seeds                                                           | $K = 3$                  | Most favorable to $K = 3$ ; a more thorough resampling variant gives 100%                                |
| Consensus_Score, net of null                                         | Algorithm co-assignment agreement, adjusted for its own null-simulation baseline (Section 3.6) | +0.311 excess over null                      | +0.149 excess over null                                                  | $K = 2$                  | Not an artifact of the score's tendency to rise mechanically with $K$                                    |
| Gap Statistic                                                        | Compares within-cluster dispersion to a uniform-reference null (Section 3.6)                   | Stopping rule not satisfied                  | Stopping rule not satisfied                                              | Neither                  | Grows monotonically through $K=6$ ; uninformative here                                                   |
| Feature-dominance resistance                                         | Share of between-cluster variance attributable to a single feature (Section 4.4)               | 82% in DPTO_FREQ alone                       | 40% in top feature (DPTO_FREQ); 3 more features add 15-28% each          | $K = 3$<br>(numerically) | Qualified below: diverse-but-scattered is not obviously preferable to dominant-but-coherent              |
| Spatial coherence of the extra cluster                               | Whether an additional cluster maps onto a stable, spatially clustered group of departments     | n/a (clean two-way split)                    | Majority label in only 2 of 26 departments; otherwise scattered by month | $K = 2$                  | $K = 2$ : clean Lima-vs-rest split; $K = 3$ 's extra cluster reads as temporal volatility, not geography |
| Temporal generalization                                              | Whether a candidate $K$ reproduces itself on the 2025 holdout without refitting (Section 4.9)  | ARI = 1.0000                                 | ARI = 0.9851                                                             | Tie                      | Both generalize well; non-discriminating                                                                 |

#### 4.3 What does the selected ( $K = 2$ ) solution actually look like?

The final  $K = 2$  model (Silhouette = 0.4496, Davies-Bouldin = 0.9235, Calinski-Harabasz = 41,817.4) separates the training data into a majority cluster (50,730 records, 77.0%) and a minority cluster (15,143 records, 23.0%) that

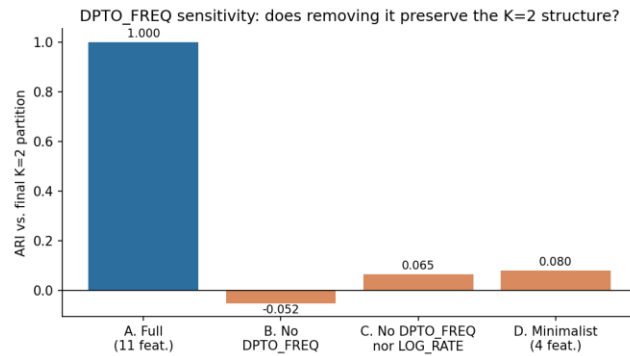
corresponds exactly to Lima Metropolitana: every Lima Metropolitana record falls in the minority cluster and no other department's records do (100% precision and recall against department identity — a perfect correspondence with a known administrative label, not a prediction evaluated against an individual-level outcome). Figure 6 decomposes exactly how much of the separation between these two clusters each feature accounts for, consistent with the dominance analysis that follows.



**Fig. 6.** Share of the total between-cluster eta-squared (variance explained) contributed by each of the eleven clustering features,  $K = 2$ , ranked largest to smallest with a cumulative-share line (right axis). The inset zooms in on every feature except DPTO\_FREQ, which would otherwise be too small to read on the main axis: LOG\_RATE\_10K, LOG\_CANTIDAD and EDAD\_ORD contribute modestly, and the remaining seven features contribute a share statistically indistinguishable from zero.

#### 4.4 What single feature is responsible for the separation, and does the structure survive its removal?

This is the central robustness result of the paper, because a partition that only exists because of one feature and a partition that reflects several independent dimensions of the data call for different scientific claims. An eta-squared decomposition attributes 82.0% of the total between-cluster variance in the  $K = 2$  solution to a single feature, DPTO\_FREQ (Section 3.3); Table 5 and Figure 4 report what happens when that dependency is tested directly rather than asserted. Removing DPTO\_FREQ alone and re-clustering at  $K = 2$  yields a partition with adjusted Rand index (ARI) = -0.052 against the original, full-feature partition — statistically indistinguishable from a random relabeling, not merely a weaker version of the same structure. This holds whether LOG\_RATE\_10K is also removed (ARI = 0.065) or the feature set is reduced to four variables with no territorial variable at all (ARI = 0.080); Silhouette remains moderate to good throughout (0.35-0.54), so some two-cluster structure persists in every reduced space — simply a different, uncorrelated one, driven in the minimal case by MONTHLY\_ANOMALY and LOG\_CANTIDAD rather than department identity. The direct conclusion is that the  $K = 2$  finding, as specified, is a function of territorial case-count concentration, and this dependence is reported here as a defining property of the result, to be explained rather than a threat to be explained away.



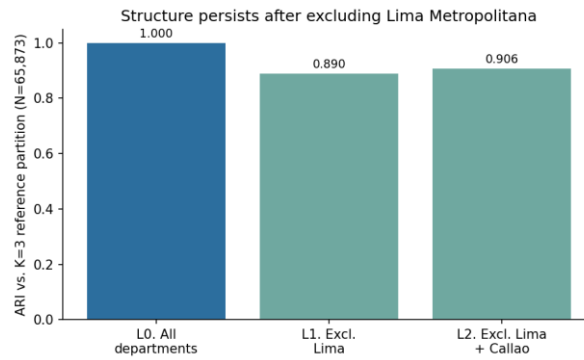
**Fig. 4.** Adjusted Rand index between the final  $K = 2$  partition and a re-clustering of the same records after removing candidate territorial/volume features, at  $K = 2$  throughout. The near-total drop from 1.000 (full feature set) to -0.052 once DPTO\_FREQ is removed — regardless of which other features are removed alongside it — is the paper's central evidence that the clustering result depends on this one feature.

**Table 5. DPTO\_FREQ sensitivity: re-clustering ( $K = 2$ ) under progressively reduced feature sets, compared by ARI against the final full-feature partition.**

| Scenario                             | Features | Silhouette | ARI vs. final $K = 2$ |
|--------------------------------------|----------|------------|-----------------------|
| A. Full feature set                  | 11       | 0.4496     | 1.000 (reference)     |
| B. Without DPTO_FREQ                 | 10       | 0.4425     | -0.052                |
| C. Without DPTO_FREQ, LOG_RATE_10K   | 9        | 0.3484     | 0.065                 |
| D. Minimal (no territorial variable) | 4        | 0.5411     | 0.080                 |

#### 4.5 Does the structure depend entirely on Lima Metropolitana specifically?

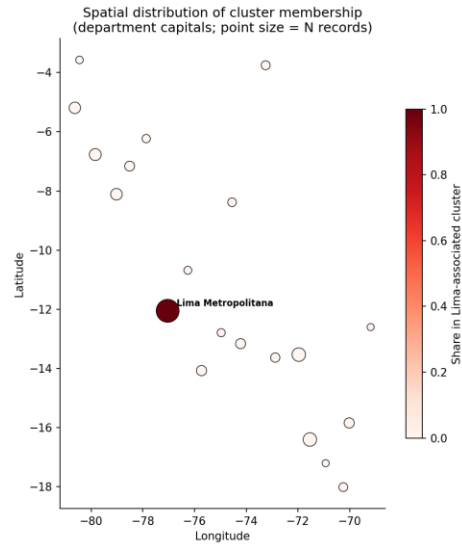
Excluding Lima Metropolitana and re-clustering the remaining 25 departments ( $N = 50,730$ ) yields  $ARI = 0.890$  against a  $K = 3$  reference partition fit on the full data; additionally excluding the contiguous department of Callao raises this to  $ARI = 0.906$  (Figure 5;  $K = 3$ , not  $K = 2$ , is used as the reference specifically for this comparison, for the reason given in Section 3.8). Both values indicate that most of the non-Lima structure captured by the reference partition survives Lima's removal, which is the complementary finding to Section 4.4: the clustering is not solely a Lima-detection exercise, even though Lima Metropolitana is, by construction, the single department with the largest effect on it.



**Fig. 5. Adjusted Rand index against a  $K = 3$  reference partition, before and after excluding Lima Metropolitana (and, additionally, Callao) from re-clustering. Values well above zero indicate that most of the non-Lima structure persists without Lima Metropolitana.**

#### 4.6 Is the structure spatially coherent, or could it be an artifact of a badly behaved spatial statistic?

Global spatial autocorrelation was tested on department-level aggregates ( $K = 4$  nearest-neighbor weights over capital coordinates, 499 permutations); Table 7 reports the results for all three variables tested. Moran's  $I$  and Geary's  $C$  show no significant global autocorrelation for the minority-cluster share, DPTO\_FREQ, or record volume. A third statistic, Getis-Ord  $G$ , could not be validly computed for the minority-cluster-share variable specifically: 20 of the 21 departments with available coordinates have a share of exactly zero (only Lima Metropolitana is non-zero), so the statistic's denominator — the sum of all off-diagonal cross-products of the variable with itself — is identically zero, and  $G$  is mathematically undefined ( $0/0$ ), a degeneracy specific to near-constant indicator-style variables. A naive implementation's permutation test can still return a small  $p$ -value in this situation (e.g.,  $p = 0.002$ ) because comparing simulated statistics against an undefined value evaluates as "false" for every comparison, mechanically producing the smallest  $p$ -value the permutation count allows; this is a numerical artifact, not evidence of spatial autocorrelation, and is excluded from the inferential argument here. Getis-Ord  $G$  is well-defined and non-significant for the two variables that lack this near-constant structure (Table 7). The spatial conclusion — no statistically significant global autocorrelation under this  $K = 4$  nearest-neighbor specification, consistent with a two-group rather than a continuous-gradient structure — rests on Moran's  $I$  and Geary's  $C$ , valid for all three variables (Figure 7); this rules out a smooth national gradient but does not, by itself, rule out finer-grained local spatial structure that a global statistic is not designed to detect.



**Fig. 7. Department-capital locations, colored by each department's share of records assigned to the Lima-associated (minority) cluster; point size encodes each department's total record volume. Lima Metropolitana (dark red) is the sole high-share outlier, with no gradient toward nearby departments.**

Spatial block cross-validation reinforces the same point from a different angle: with DPTO\_FREQ included, only 1 of 6 geographic blocks contains both clusters (the other five are single-cluster and cannot support a within-block validity estimate, because Lima Metropolitana falls entirely within one block); without DPTO\_FREQ, all 6 of 6 blocks contain both clusters. This identifies the specific cause of the spatial blocks' lack of cluster diversity: four of six blocks fall entirely within a single cluster once DPTO\_FREQ is included.

#### 4.7 Are the reported associations with socioeconomic covariates real, or an artifact of counting the same department many times?

An external validation against four departmental covariates (2021 Human Development Index, poverty rate, urbanization rate, population density) depends entirely on the unit of analysis used to test it (Table 6). At the individual-record level ( $N = 65,873$ , each department's four covariate values repeated once for every record from that department), all four covariates are "significant" at  $p < 0.001$  by a Kruskal-Wallis test (Kruskal and Wallis, 1952); this is a textbook pseudoreplication artifact, because repeating the same four departmental numbers across thousands of rows from the same department does not create thousands of independent observations — the covariates in fact carry only 25 independent values, one per department. At the correct department-level unit ( $N = 25$  descriptively, or  $N = 24$  under the  $K = 3$  grouping used for the significance test specifically, since Lima forms a group of size one under  $K = 2$  and a two-group test is invalid with  $n = 1$ ), none of the four covariates reach significance ( $p = 0.076$ - $0.296$ ). A descriptive contrast, not a fabricated p-value, illustrates the same point at  $K = 2$ : Lima's 2021 HDI sits at the 100th percentile of the other 24 departments, its poverty rate at the 25th, and its urbanization and population density both at the 95.8th — a department that is simultaneously unusual and ordinary depending on which covariate is examined, which is not the profile a single coherent socioeconomic mechanism would be expected to produce.

**Table 6. Kruskal-Wallis test of four socioeconomic covariates against cluster membership, at the pseudoreplicated individual-record unit vs. the correct department-level unit. None reach significance at the correct unit.**

| Covariate          | Individual-level<br>( $N = 65,873$ ) $p$ | Dept-level, $K = 3$<br>grouping ( $N = 24$ ) $p$ |
|--------------------|------------------------------------------|--------------------------------------------------|
| 2021 HDI           | <0.001                                   | 0.076                                            |
| Poverty rate       | <0.001                                   | 0.076                                            |
| Urbanization rate  | <0.001                                   | 0.296                                            |
| Population density | <0.001                                   | 0.117                                            |

#### 4.8 Does the individual-level pattern survive aggregation to coarser territorial-temporal units?

Two independent aggregations were tested. Department-year cells ( $N = 156$ , exactly matching the  $26 \times 6$  theoretical department  $\times$  year combinations) and department-month cells ( $N = 1,864$  of 1,872 theoretically possible, eight cells having zero records) each independently recover a two-cluster solution whose minority cluster is 100% Lima Metropolitana, and the individual-level  $K = 2$  partition is recoverable from either aggregate with  $ARI = 1.0000$ . This means the territorial-temporal aggregate is sufficient to recover the same partition the individual-level clustering produces — a useful robustness check on where the signal lives, but not an independent replication in a new setting, since both aggregates are constructed from the same underlying records as the individual-level analysis.

#### 4.9 Is the partition stable when applied to a later, held-out year?

The final  $K = 2$  model, fit only on 2019-2024 data, was applied without any refitting to the fully held-out 2025 test partition ( $N = 11,458$ ): every Lima Metropolitana record in the 2025 data falls in the Lima-associated cluster and no other department's records do (100% precision and recall against department identity, again a correspondence with a known label rather than an individual-level prediction). An independently refit  $K = 2$  model, fit only on the 2025 data in isolation, agrees with the inherited assignment perfectly ( $ARI = 1.0000$ ). Both results indicate the two-cluster structure is stable across a later calendar year not used to fit the model, rather than an artifact of the specific training window. This result should nonetheless be interpreted as a stability check on a territorially-dominated partition, not as a fully independent temporal validation: because DPTO\_FREQ (Section 3.3) is itself constructed from the full 2019-2025 span, the training and test periods are not completely isolated for that one feature, and Section 5.6 discusses this limitation directly. Consistent with Section 4.4, this is a claim about the stability of a feature-dominated partition under the pipeline's existing design, not evidence that the partition captures a multidimensional structure independent of that feature.

**Table 7. Summary of spatial, multiscale and temporal robustness checks for the final  $K = 2$  partition (Sections 4.6, 4.8 and 4.9).**

| Check                                      | What it tests                                        | Result                                            | Supports                                            |
|--------------------------------------------|------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------|
| Global spatial autocorrelation (Moran's I) | Geographic clustering beyond the Lima/rest split     | $I = -0.050$ , $p = 0.404$ (not significant)      | No continuous spatial gradient                      |
| Global spatial autocorrelation (Geary's C) | Same, alternative statistic                          | $C = 1.000$ , $p = 0.430$ (not significant)       | No continuous spatial gradient                      |
| Spatial block cross-validation             | Both clusters present in each of 6 blocks?           | 1 of 6 blocks with DPTO_FREQ; 6 of 6 without it   | Isolates DPTO_FREQ as cause of block-level scarcity |
| Department-year aggregation                | Individual partition recoverable from $N=156$ cells? | $ARI = 1.0000$                                    | Signal concentrated in aggregate structure          |
| Department-month aggregation               | Same, at finer $N=1,864$ -cell resolution            | $ARI = 1.0000$                                    | Confirms department-year result, finer resolution   |
| Temporal generalization, inherited model   | 2019-2024 model recovers Lima in unseen 2025?        | 100% precision/recall against department identity | Partition is stable across a genuinely unseen year  |
| Temporal generalization, independent refit | 2025-only refit agrees with inherited assignment?    | $ARI = 1.0000$                                    | Stability is not an artifact of the training window |

## 5. Discussion

### 5.1 Main finding: stability does not imply multidimensionality

RENIPED separates, consistently across six clustering algorithms, into a Lima-Metropolitana group and a rest-of-Peru group. That separation is genuine in the narrow sense that it survives six different algorithmic assumptions and an unseen year, but it is not evidence of a multidimensional territorial-temporal profile in the sense the original pipeline's framing implied. One engineered feature, DPTO\_FREQ, is by construction a near-direct encoding of “is this Lima Metropolitana,” and it carries 82.0% of the variance that separates the two clusters (Section 4.4); once it is removed, a different and statistically unrelated two-cluster structure emerges from the remaining features ( $ARI \approx 0$  against the

original). Stated plainly, the central finding is that a clustering solution can be highly stable, reproducible across algorithms, and stable across an unseen year, while remaining scientifically limited by the dominance of a single administrative feature: stability of this kind is not, by itself, evidence of multidimensionality.

### 5.2 Implications for GeoAI and administrative data

A raw case count is mechanically larger where more people live and where reporting infrastructure is stronger, and Lima concentrates both; the clustering procedure did not manufacture this pattern, it found the pattern that the feature set makes easiest to find. This is a specific instance of a broader risk for GeoAI applications to administrative registries generally: a count-based feature that mechanically tracks jurisdiction size or reporting capacity can dominate an unsupervised result in a way that looks like discovery but is not. The individual-level literature on missing persons (James et al., 2008; Jeanis et al., 2020; Enríquez et al., 2024) and the two prior applications of clustering to missing-persons data specifically (Johnson et al., 2025; Pierzchała and Gutowski, 2021) do not test whether a registry-level partition is a multidimensional structure or a single-feature artifact, because that question sits outside each of their respective scopes; this paper’s results are consistent with, and give a concrete mechanism for, the broader methodological concern documented elsewhere for spatial and epidemiological clustering generally (Auchincloss et al., 2012; VoPham et al., 2018).

### 5.3 Implications for missing-person analytics

Read narrowly, this bounds what the partition can be used for. At  $K = 2$ , the partition’s known DPTO\_FREQ dependence means it should be read as “Lima Metropolitana is disproportionately represented in this registry,” not as risk varying smoothly across the rest of the country’s geography, which the non-significant global spatial-autocorrelation results likewise do not support (Section 4.6). Three further claims are not supported and are explicitly disclaimed here. The clustering does not demonstrate a validated socioeconomic mechanism: the individual-level significance test that appeared to show one is a pseudoreplication artifact, and the correctly specified department-level test finds no significant association for any of the four covariates tested (Section 4.7). It does not demonstrate individual-level risk: every feature in this study is recorded at the level of a report, a department, or a calendar period, never at the level of a person’s individual characteristics or circumstances, so no claim about which individuals are more likely to go missing follows from any result here. And generalization to the 2025 holdout (Section 4.9) is evidence of stability for this territorially-dominated partition specifically, not a claim that the same feature-dominance pattern would appear in a different country, a different registry design, or a differently structured administrative dataset.

### 5.4 Methodological implications

Any downstream use requiring certainty about a distinct third group should not proceed without independently confirming the department-level pattern behind  $K = 3$ , since this paper’s own spatial-coherence check (Section 4.6) found it wanting. More generally, the contribution here is a disclosed, reproducible template for five things every registry-clustering study of this kind faces but few report in full: that feature dominance can be quantified rather than assumed or denied outright (Section 4.4); that a close cluster-count decision is properly a multi-criteria trade-off, not a single convenient number (Table 4); that a multiscale robustness check and an individual-level validation answer different questions, so recovering a partition from department-year or department-month aggregates (Section 4.8) is evidence of the former, not an independent replication in a new setting; that a unit-of-analysis error can manufacture an apparently strong external validation where none exists (Section 4.7); and, drawing these together, that a highly reproducible partition does not by itself imply a multidimensional one.

### 5.5 Generalizability

The five points above transfer in principle to other national administrative registries with comparable territorial concentration in case reporting, wherever a raw count feature tracks jurisdiction size or reporting capacity. What does not generalize without further work is the empirical finding itself: that RENIPED’s partition is dominated by DPTO\_FREQ to the degree measured here (82.0%). Whether a comparably dominant feature exists in another country’s registry is a question this paper’s method could answer elsewhere, but that its results alone do not.

### 5.6 Limitations and future work

The limitations below are attached to the specific analysis each qualifies. DPTO\_FREQ does not achieve complete temporal isolation between training and test (Section 3.3): it is built from the full 2019-2025 span, so every record’s value already reflects information from the 2025 year later used as the holdout. No version of this pipeline recomputing DPTO\_FREQ from the 2019-2024 training years alone has been executed, so the size of any resulting effect is not quantified here and is not treated as ruled out; a training-only recomputation of DPTO\_FREQ is identified as a priority for a future revision. Accordingly, the 2025 result in Section 4.9 is read as an evaluation of the stability of a territorially-dominated partition under the pipeline’s existing design, not as a fully independent temporal validation free of this construction choice.

Department population and socioeconomic covariate figures are single, time-invariant cross-sections, not a year-by-year panel, so the department-level test in Section 4.7 is a single comparison rather than a longitudinal one. Spatial statistics use department-capital coordinates rather than population-weighted centroids; a different coordinate choice would not resolve the Getis-Ord degeneracy in Section 4.6, which follows from the near-constant structure of the variable itself, but could change the block cross-validation’s adjacency structure in ways not tested here. The SHAP-based explanation is one step removed from the clustering it describes (Section 3.9): it explains a Random Forest trained to imitate the cluster labels, not the clustering algorithm directly, and its near-perfect fidelity is a property of a feature-dominated partition rather than a guarantee that the explanation is complete. Most substantively, this paper does not include density-based clustering (HDBSCAN), spatial regression (SAR/SEM), a leave-one-region-out spatial validation, survival and transition-matrix analyses of case duration, and a gradient-boosted predictive component. None of these are reproduced in the notebook accompanying this paper, and their results are not reported here; they are named only as directions for future extension, not as completed contributions of this study.

## 6. Conclusions

A national-scale missing-persons clustering pipeline was re-executed from freshly downloaded source data to test, rather than defend, its own prior conclusions. A two-cluster partition separating Lima Metropolitana from the rest of Peru is recovered consistently across six clustering algorithms (mean pairwise ARI = 1.000) and remains stable under this pipeline’s design when applied to a held-out year (ARI = 1.0000), but is also shown to depend on a single territorial-frequency feature for 82.0% of its separating variance — a dependency this paper reports as a defining property of the result, not a threat to it. A three-cluster alternative, favored by one of eight decision criteria, is retained as a secondary finding best explained as a temporal-volatility regime rather than a third territorial category. The apparent socioeconomic association is shown to be a pseudoreplication artifact that does not survive correction to the correct unit of analysis. The main contribution is methodological: a disclosed, reproducible procedure for distinguishing genuine multidimensional structure from single-feature dominance in administrative clustering — stability is not, by itself, evidence of multidimensionality — applicable beyond this specific registry. The main limitation is scope: these results describe reporting patterns in one national registry over one period, using a territorial-frequency feature that is not fully isolated from the test period, and should not be read as a general claim about the geography of missing-persons risk, in Peru or elsewhere.

## Declarations

**Ethics statement.** This study relies exclusively on de-identified administrative microdata from RENIPED — department, province, district, sex, age bracket, nationality, year and month per record — with no names or other direct identifiers accessed at any stage of the analysis. No primary data were collected from human participants; the department-level socioeconomic and population figures used for external validation are drawn from publicly published national statistics (Section 3.1, Section 4.8).

**Funding.** This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

**Declaration of competing interest.** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

**CRedit authorship contribution statement.** Moisés Evangelista Gamarra: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. Edwar Abrahan Alama Carreño: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

## Data Availability

The missing-persons registry export used in this study is derived from Peru’s National Registry of Information on Missing Persons (RENIPED) (Peruvian National Police, n.d.) and the national open-data platform (National Open Data Platform, n.d.); the specific CSV export analyzed here, together with its SHA-256 content hash, is available at the location documented in the accompanying reproducibility notebook. Department-level population and socioeconomic covariate figures (2021 cross-section) are reported in full in the accompanying notebook’s source code (Section 3.1, Section 4.8).

## Code Availability

All code used to produce every number, table and figure in this paper is provided as a single executed notebook, C20-2026\_V4\_FINAL\_REPRODUCIBLE.ipynb, including random seeds (42 for all clustering; seed-indexed for all resampling procedures), exact library versions (Python 3.12.3; numpy 2.4.4; pandas 3.0.2; scikit-learn 1.8.0; scipy

1.17.1; shap 0.52.0; libpysal 4.15.0; esda 2.10.0), and the full feature-engineering, model-selection, sensitivity, spatial, temporal and explainability pipeline described in Section 3. HDBSCAN, spatial-regression, LORO, survival/transition-matrix and gradient-boosting code referenced in Section 5.6 as future extensions are explicitly not included, as noted there.

## References

- Amnesty International Peru. (2025). Cusco continues to face alarming levels of gender-based violence.
- Anselin, L. (1995). Local indicators of spatial association—LISA. *Geographical Analysis*, 27(2), 93–115. <https://doi.org/10.1111/j.1538-4632.1995.tb00338.x>
- Auchincloss, A. H., Gebreab, S. Y., Mair, C., & Diez Roux, A. V. (2012). A review of spatial methods in epidemiology, 2000–2010. *Annual Review of Public Health*, 33, 107–122. <https://doi.org/10.1146/annurev-publhealth-031811-124655>
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Caliński, T., & Harabasz, J. (1974). A dendrite method for cluster analysis. *Communications in Statistics*, 3(1), 1–27. <https://doi.org/10.1080/03610927408827101>
- Carvajal, A. E. (2020). COVID-19 and the limitations of official responses to gender-based violence in Latin America: Evidence from Ecuador. *Bulletin of Latin American Research*, 39(S1), 7–11. <https://doi.org/10.1111/blr.13188>
- Davies, D. L., & Bouldin, D. W. (1979). A cluster separation measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-1(2), 224–227. <https://doi.org/10.1109/TPAMI.1979.4766909>
- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26. <https://doi.org/10.1214/aos/1176344552>
- Enríquez, Y., Cahui, C. R., & Díaz, G. M. (2024). An ecological approach to the characteristics and determinants of sexual violence against adolescents in Peru. *Biomédica*, 44. <https://doi.org/10.7705/biomedica.7131>
- Ferguson, L., & Picknell, W. (2021). Repeat or chronic? Examining police data accuracy across the “history” classifications of missing person cases. *Policing and Society*, 1–15.
- Getis, A., & Ord, J. K. (1992). The analysis of spatial association by use of distance statistics. *Geographical Analysis*, 24(3), 189–206. <https://doi.org/10.1111/j.1538-4632.1992.tb00261.x>
- Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1), 193–218. <https://doi.org/10.1007/BF01908075>
- Infobae. (2025a, January 30). Gender-based violence in Peru: Femicides and disappearances reach alarming levels.
- Infobae. (2025b, August 2). Alarming increase in femicides and disappearances in Peru: Figures rise 11% in 2025, Ombudsman’s Office warns.
- International Commission on Missing Persons (ICMP). (2024). Annual report 2024. [https://icmp.int/wp-content/uploads/2025/06/250415\\_ICMP\\_ANNUAL-REPORT-2024\\_DIGITAL\\_DEF-5.pdf](https://icmp.int/wp-content/uploads/2025/06/250415_ICMP_ANNUAL-REPORT-2024_DIGITAL_DEF-5.pdf)
- International Committee of the Red Cross (ICRC). (2024). ICRC strategy 2024–2027. <https://www.icrc.org/en/publication/4745-icrc-strategy-2024-2027>
- James, M., Anderson, J., & Putt, J. (2008). Missing persons in Australia (Research and Public Policy Series No. 86). Australian Institute of Criminology.
- Janowicz, K., Gao, S., McKenzie, G., Hu, Y., & Bhaduri, B. (2020). GeoAI: Spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond. *International Journal of Geographical Information Science*, 34(4), 625–636. <https://doi.org/10.1080/13658816.2019.1684500>
- Jeanis, M. N., Fox, B. H., Jennings, W. G., Perkins, R. C., & Liberto, A. (2020). Oooh she’s a little runaway: Examining invariance in runaway youth trajectories by developmental and life-course risk factors and gender. *Journal of Developmental and Life-Course Criminology*, 6(4), 398–423.
- Johnson, L. J., Listi, G. A., Wilson, T. V., & Leitner, M. (2025). A spatial clustering analysis of missing persons data to generate spatially informed community outreach event locations. In *New Research in Crime Modeling and Mapping Using Geospatial Technologies*. Springer. [https://doi.org/10.1007/978-3-031-81580-5\\_14](https://doi.org/10.1007/978-3-031-81580-5_14)
- Kruskal, W. H., & Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47(260), 583–621. <https://doi.org/10.1080/01621459.1952.10483441>
- Li, W. (2020). GeoAI: Where machine learning and big data converge in GIScience. *Journal of Spatial Information Science*, (20), 71–77. <https://doi.org/10.5311/JOSIS.2020.20.658>

- Li, W., Arundel, S. T., Gao, S., Goodchild, M. F., Hu, Y., Wang, S., & Zipf, A. (2024). GeoAI for science and the science of GeoAI. *Journal of Spatial Information Science*, (29), 1–17. <https://doi.org/10.5311/JOSIS.2024.29.349>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.
- Moran, P. A. P. (1950). Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2), 17–23. <https://doi.org/10.1093/biomet/37.1-2.17>
- NACLA Report on the Americas. (2025, July). Feminicidal disappearance in Peru.
- National Open Data Platform, Government of Peru. (n.d.). Missing persons. <https://www.datosabiertos.gob.pe/dataset/personas-desaparecidas>
- Ord, J. K., & Getis, A. (1995). Local spatial autocorrelation statistics: Distributional issues and an application. *Geographical Analysis*, 27(4), 286–306. <https://doi.org/10.1111/j.1538-4632.1995.tb00912.x>
- Pediatrics. (2020). Runaway youth: Caring for the nation’s largest segment of missing children. *Pediatrics*, 145(2), e20193752. <https://doi.org/10.1542/peds.2019-3752>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Perez-Vincent, S. M., & Carreras, E. (2022). Domestic violence reporting during the COVID-19 pandemic: Evidence from Latin America. *Review of Economics of the Household*, 20(3), 799–830. <https://doi.org/10.1007/s11150-022-09607-9>
- Peruvian National Police. (n.d.). National Registry of Information on Missing Persons (RENIPED). <https://desaparecidosenperu.policia.gob.pe/Desaparecidos/reniped>
- Peruvian Ombudsman’s Office. (2025, February). What happened to them? Equality and Non-Violence Report No. 61. <https://www.defensoria.gob.pe/wp-content/uploads/2025/03/Reporte-Igualdad-y-No-Violencia-N-61-Qué-pasó-con-ellas-Febrero-2025.pdf>
- Pierzchała, D., & Gutowski, T. (2021). Machine learning-based method for recommendation of missing person’s “search level.” In *Proceedings of the 37th International Business Information Management Association Conference (IBIMA)*.
- Quinteros, V. M. (2025, April 30). How many are there and where are they? Missing persons and a search without answers. IDEHPUCP.
- Quinteros, V. M. (2026, January). Missing in 2025: A blind search and an alarming rise in reports. IDEHPUCP.
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
- Ruiz Reyes, J., Congram, D., Sirbu, R. A., & Floridi, L. (2025). Where are they? A review of statistical techniques and data analysis to support the search for missing persons and the new field of data-based disappearance analysis. *Forensic Science International*, 376, Article 112582. <https://doi.org/10.1016/j.forsciint.2025.112582>
- Tibshirani, R., Walther, G., & Hastie, T. (2001). Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B*, 63(2), 411–423. <https://doi.org/10.1111/1467-9868.00293>
- VoPham, T., Hart, J. E., Laden, F., & Chiang, Y.-Y. (2018). Emerging trends in geospatial artificial intelligence (geoAI): Potential applications for environmental epidemiology. *Environmental Health*, 17(1), 40. <https://doi.org/10.1186/s12940-018-0386-x>
- World Bank. (2022, March). The incidence of gender-based violence in Latin America and the Caribbean.